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Are Pairs More Attentive Towards Feedback Than Individuals? A Glance Into Feedback Neglect for Middle-School Students Using an Educational Game in History

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Abstract: Critical constructive feedback (CCF), is said to be a key ingredient in learning. However, if and how learners engage in CCF is rarely studied in collaborative learning situations. This study aims to filling this research gap by analyzing data logs from dyads of middle school students using an educational game in history, evaluating their inclination to dismiss or attend to feedback when making errors and failing on tasks. Data from 106 students, working individually as well as in pairs on the same tablet, were analyzed and evaluated. The results reveal that the inclination to attend to feedback differed between conditions, but that this tendency also differed between achievement levels. While medium/lower-achieving students generally were more attentive towards feedback when working in pairs (compared to working individually), the opposite was found for higher-achieving students. However, higher-achieving students working in pairs attended more to CCF after repeated failures (compared to working individually).

Introduction

Research on collaborative learning (CL) often argues that social interaction not only scaffolds learners to better understand a problem, but that it also may increase students' self-esteem and engagement (Lai, 2011). When solving problems in pairs or in smaller groups, students are forced to share responsibilities and negotiate about answers and solutions (Schwartz & Lin, 2000). Collaboration also seems to empower self-regulation, making students more cognitively aware of their learning processes (Hadwin et al., 2017; Schoor et al., 2015). At the same time, when collaborating in a group, the risk of making errors as well as the risk of feeling insufficient might be divided between the group members, possibly making them more inclined to accept and embrace external critical constructive feedback (CCF). However, even though there is plenty of research on how individuals react to CCF (by, for instance, accepting it or dismissing it), we know less about how *groups* handle CCF – and especially so if this is delivered from a teacher or a system. The study at hand aims to address this research gap by crossing the borders between research on CCF and research on CL, analyzing a specific type of behavior, e. g. feedback neglect, for students working in pairs and individually. We thereby hope to not only contribute to the understanding of CL, but also to explore new ways of evaluating feedback engagement in a common collaborative learning situation: students playing an educational game via a shared tablet or laptop in the classroom.

Collaboration and theories on productive agency

What do we mean by collaborative learning? Ever since Dillenbourg (1999) asked the question – and made a commendable effort to respond to it – the research community defining the concept of CL has expanded and diversified. Consequently, a researchers' view of CL will heavily influence not only how to conduct experiments and/or observations, but also how to best measure the effects of CL, and what conclusions to draw about the reasons behind these effects.

According to Schwartz and Lin (2000), the primary driving force in collaborative learning is each participant's *productive agency*. Thus, simply “working together” is not necessarily a collaborative process. In order to cooperate, however, “individuals have to enter into relationships, to produce ideas, to choose whether to communicate, and they have to choose whether to compromise their goals” (Schwartz & Lin, 2000, p. 8). Consequently, group members need to share meaning and effort and represent each other's thoughts, a process that can be both time-consuming and ineffective. Effectiveness or outcomes on a group level are thereby not sufficient criteria for evaluating collaboration. Instead, it would be more appropriate to evaluate the frequency of rich interactions (Dillenbourg et. al, 2016; Webb, 1991). Dillenbourg (1999) also emphasizes that even if some behaviors or processes can be scaffolded through collaboration, the underlying mechanisms involved in CL have to be the same as the ones involved in individual cognition – at least at some basic level. While recognizing that there are other useful perspectives on CL, the present study takes the stance that observed behavioral patterns (in this case, attention towards feedback) can be interpreted as originating from individual states and minds, even

though they may be mediated through collaboration with others. By embracing this view, it becomes possible to compare one student's behavior in one setting (working alone) with the same student's behavior in another setting (working in pairs) by the means of a within-subjects design.

Working in pairs on tablets

The research field of computer supported collaborative learning (CSCL) is wide and heterogeneous. For instance, while there is a broad understanding of the strength and potential with CSCL in a wider context, there is much less in-depth knowledge about how students cooperate during tablet-based small-group activities in ordinary classrooms (Fleck et al., 2021). Mobile devices are, however, often used in quite loose constellations in classroom settings, where they may lead to informal collaboration (Geer et al., 2017), and it is not uncommon for teachers to let students share tablets when using Edtech apps (Fleck et al., 2021). Furthermore, even if several studies show that collaboration on tablets may be beneficial for open-ended applications (Falloon & Khoo, 2014), to our knowledge, there are very few studies evaluating the effects of pairs using them for drill- and practice activities.

The research on students' learning with tablets also appears to be divided. Studies targeting individual use of tablets or computers in schools – and especially in K-12 education – often make use of quantitative learning outcomes based on surveys and/or large-scale log data. Research focusing on cooperation around shared mobile devices, on the other hand, is more often based on observations, interviews and/or registration of verbal and non-verbal communication (see for instance Davidsen & Vanderlinde, 2016). The underlying reason for this methodological split is not mysterious, quite the opposite: While individual use of digital devices produces vast amount of data traces but few observable behaviors, collaborative work often result in lively human-to-human interaction. At the same time, data logs emerging from collaborative interactions can be hard to deduce. Do they the result from the work of one the peers in the group or from all of them? However, if we can't use the same method for investigating individual and cooperative behaviors, how do we know if a student changes his/her behavior when shifting from one learning situation to another?

In the present study, we have chosen a quantitative approach, evaluating only data traces. By doing so, we assume that these traces, when they derive from pairs, originate from both students simultaneously, although we cannot be totally sure of it. Naturally, it would have been beneficial to complement these logs with observations or recordings. However, since the study was conducted remotely due to the Covid19-pandemic, and the students participated in larger groups (full classes), registering conversations or other human-to-human-interactions would have been extremely difficult. On the other hand, if differences in data logs (from individuals or pairs) can be modelled in a satisfying way, resulting in statistically significant results, we might claim this method to be valid.

The problem with feedback neglect – and the possibilities to decrease it

Whatever the theoretical standpoint may be for optimizing teaching and learning situations, feedback is generally considered a key element in the learning process (Hattie & Timperley, 2007; Shute, 2008). Embracing feedback is not only important for correcting mistakes, but it is also an indication of self-regulation and self-awareness (Cutumisu et al., 2015). A delicate problem is, however, that students often tend to ignore feedback, and especially if its negative (Hattie & Timperley, 2007). Recent studies on feedback in educational games reveal that students sometimes neglect as much as half of the critical constructive feedback messages (CCFs) (Segedy et al., 2012; Tärning et al., 2020). The reasons for neglecting CCF can be several. Feelings of personal failure, a sense of being punished, and thoughts about personal incapability are just some of the emotional responses that can be evoked by negative criticism (Kluger & DeNisi, 1996). However, it seems that students are more resilient towards failure, and hence also more positive towards engaging in feedback, if they are tutoring others – such as less knowledgeable peers or virtual teachable agents (Silvarg et al., 2020; Tärning et al., 2020). The engagement in a (virtual) companion here appears to not only trigger a sense of responsibility towards that companion, but also function as an ego-protective buffer (Chase, 2011). This in turn strengthens the individual to act on the companions' behalf – i.e., to engage in critique and feedback to increase performance.

In a CL situation, peers can not only give each other CCF, but they can also receive CCF together as a pair. As of today, the former is more thoroughly investigated and there is a growing body of research evaluating regulation in collaborative learning environments (Hadwin et al., 2017; Schoor et al., 2015). How group members react to *external* CCF is, however, not well known. Since the collaborative setups described above (where peers teach virtual agents or one another) evidently can reduce feedback neglect, we might assume that CL, as well as CSCL, may do the same. Consequently, the study at hand focused on the probability of students accepting CCF when failing on tasks in an educational game, and if this acceptance differed if the students played for themselves or in pairs. More specifically, our research questions were: i) Do students playing the game in pairs attend to feedback more frequently than if they play for themselves? ii) Is this behavior affected by severe and/or repeated failing? and iii) Does the feedback behavior differ between higher- and lower-achieving students?

Method

To address the research questions, we designed a quasi-experimental study with repeated measures within groups, where middle-school students used an educational game in class with their regular teachers present to help and guide them. The experiment was conducted in Sweden in spring 2021, during the Covid19-pandemic. It was therefore adapted to adhere with the recommendations of the Public Health Agency in Sweden (Folkhälsomyndigheten in Swedish), which, at the time, allowed students to attend schools but advised citizens to minimize interactions with others (HSLF-FS 2020:80, Folkhälsomyndigheten). Consequently, classrooms were not visited for data collection and all forms of coordination with the teachers were digitally mediated.

The educational game

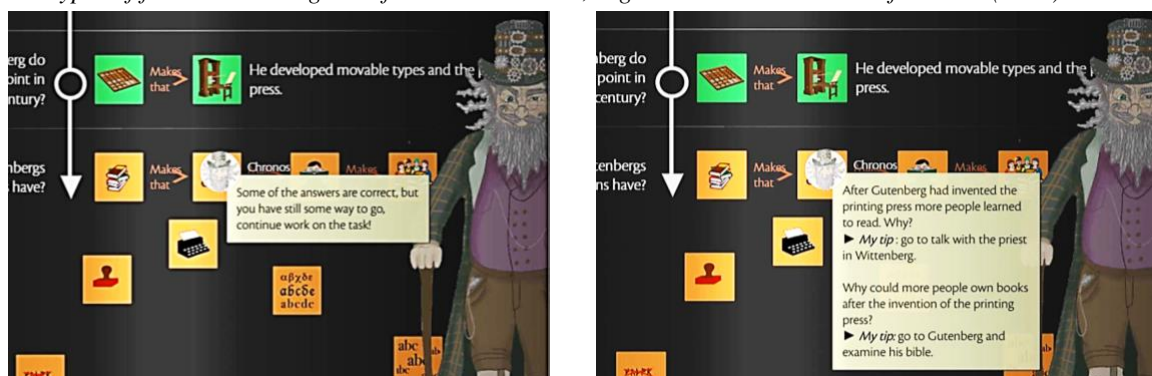
The experimental stimulus consisted of an educational game in history that has been used in a series of research projects on feedback engagement during recent years (Silvervarg et al., 2020; Ternblad & Tärning, 2020; Tärning et al., 2020). The students playing the game travel through the history of science, visit historical scientists, search for information, and solve tasks (on the format of concept maps, timelines, multiple-choice questions etc.). To progress in the game, each task must be completed before starting on the next one. After presenting a solution to a task, the students receive feedback in two parts: i) a general task assessment, and ii) critical constructive feedback on errors together with suggestions about how to address them (Figure 1). Depending on the level of correctness, the first part is formulated as follows:

- 100% correct (Passed): “The task is approved, everything is correct. Great work!”
- 100-80% correct (Passed): “Only some minor error. The task is approved. Great work!”
- 75-80% correct (Failed but ‘Almost Correct’): “The major part is correct, great work!”
- 60-74% correct (Failed and ‘Quite Correct’): “A fair amount is correct, not far to go now, great work!”
- 30-59% correct (Failed and ‘Quite Wrong’): “Some things are correct, but you gave still some way to go, continue to work on the task!”
- <30% correct (Failed and ‘Very Wrong’): “Unfortunately, a lot is missing or wrong. Keep on working!”

After this assessment, the student clicks to proceed. The subsequent CCF provides information about some of the errors and presents hints. It is structured as follows: “Some facts concerning Mr/Mrs X’s inventions regarding A are not correct. Travel back and speak to him/her, locate hi/hers book on the shelf behind him/her and read about B.” The feedback messages vary in phrasing and content depending on the errors made, and the student can choose to read them, dismiss them (by clicking them down before paying any attention to them) and/or act upon them or not. After reading (or ignoring) the CCF, the student clicks to proceed – and either just tries to solve the task again or makes a new time travel to gather new information. Notably, the game is not explicitly designed for collaboration and as of today, only individual use of the game has been studied.

Figure 1

Two types of feedback messages. Left: task assessment, Right: critical constructive feedback (CCF)



Participants and procedure

209 students from 6 classes in 5th and 6th grade (10-12-year-olds) in southern Sweden were recruited for the experiment. Since self-regulation and feedback engagement is known to rely heavily on academic performance, the teachers were asked to rate the children’s achievement levels (based on their reading comprehension) on a

three-dimensional scale (higher, medium, and lower) prior to the experiment. The teachers then helped us to match students with similar achievement levels into suitable homogeneous pairs. It should be noted that we are very well aware of that research on CL often recommend heterogeneous groups, where lower-achieving students may learn from more capable peers (Lai, 2011). However, some researchers also point out that symmetry can be beneficial for CL, not at least to assure all group members equal status and equal opportunities to contribute (Kozlov & Große, 2016). In our specific case, the homogeneous pairs made it possible for us to hypothesize that the observed behavior derived from both participants in a democratic manner.

In the actual intervention, the students used the game in class at two separate occasions, 35-40 minutes at a time. At one of the two sessions, the students worked individually with one tablet, and at the other they shared the tablet with their matched classmate. To avoid order-effects, the order (single-to-pair or pair-to-single) was randomly assigned to each class. The first session began with a short introduction given by the experiment leader via video link, followed by a film clarifying the narrative of the game. Teachers, and in some cases assisting resource personnel, helped out with the distribution of links and log-in details. The students then played the game for the rest of the session. The first session contained three tasks with different historical events and scientists (Copernicus, Galilei, and Newton), and the second session consisted of three new tasks (Châtelet, Gutenberg and a task mixing information from several historical events). The difficulty level on the tasks varied, with Copernicus being the easiest and Newton and Gutenberg being more difficult. If students had any questions during the gameplay, they were instructed to approach the video link and talk to the researcher in charge. Teachers were instructed to assist with technical issues if needed, but to avoid helping students to solve the tasks. Since all students were using the game at their own pace, some of them finished all tasks before the end of each session, while others didn't finish any. The actual contribution to the data set from each student does thereby vary.

Measures and data gathering

Time stamped data logs were gathered for all students' individual logins. Pairs only used one single login, and this data log was duplicated to apply for both students. The students' attention towards feedback was estimated by measuring the amount of time the feedback message was displayed on screen. In a previous study using the same stimulus, it was evaluated that students needed at least 4.25 seconds to grasp the content in such a message (Silvervarg et al., 2020). This corresponds to reading 10-11 words at a speed of 2.5 words/second (150 words/min). If the student(s) clicked down the CCF at an earlier stage (that is, before 4.25 seconds) the feedback was classified as ignored, otherwise attended to. Other measures used in the data logs were the aggregated number of unsuccessful trials and the task-assessment category (Almost Correct, Quite Correct, Quite Wrong and Very Wrong).

Result

In total, 209 students participated in the study. However, not all of them attended both sessions, and some of the pairs were reorganized on site and thereby difficult to track. In addition, not all of the students received CCF messages – sometimes because they did not engage in the game and never tried to solve a task, sometimes because they solved the tasks instantly and never failed. Consequently, for a student to contribute to the data set, he/she had to have received at least one CCF in both sessions. This narrow criterion led to the exclusion of a large number of logs, and the remaining final dataset thereby consisted of data from 106 students (44 boys and 62 girls). The distribution of higher-, medium-, and lower-achieving students playing single-to-pair and pair-to-single is presented in Table 1 below. As shown, the number of students differs slightly between groups.

Table 1

Distribution of students playing the game first individually, then in pairs – and the other way around

Reading Compr. Level	Single Session 1 / Pair Session 2	Pair Session 1 / Single Session 2
Lower (N=21)	10	11
Medium (N=43)	17	26
Higher (N=42)	10	32
Σ (N=106)	37	69

Since there were fewer lower-achieving students in the dataset, and since we know from previous studies that higher-achieving students often are less affected by learning conditions in general (Kluger & DeNisi, 1996; Lai, 2011; Tärning et al., 2020), we decided to merge the medium- and lower-achievers into one single category. In addition, we didn't want to include CCF's that originated from excessive repeated failing, since this could not

only provoke students to behave in an uncharacteristic way (frustrated and irrational guessing etc.), but also induce a risk for asking for help from teachers or peers. To counteract this, we introduced a threshold for how many failed trials from one single task that should be included in the analyses. We here decided on 6 trials per task and student. Hence, each student contributed to the data set with everything from 6 to 36 trials.

In average, medium- and lower-achievers playing individually ($N = 64$) contributed to the final dataset with 5.5 CCFs ($SD = 2.7$), while the higher-achievers ($N = 42$) playing individually contributed with slightly fewer CCF's ($M = 4.7$, $SD = 2.7$). Medium- and lower-achievers playing in pairs contributed with 4.6 CCF's in average ($SD = 2.4$), which is very similar to the pair-playing higher-achievers ($M = 4.6$, $SD = 2.3$).

Attention to feedback when playing in pairs or playing individually

The main research question addressed students' inclination to attend to CCF in the two different conditions (using the game individually or together with a peer). As attention to feedback can be related to in-game performance (Ternblad & Tärning, 2020), it was also hypothesized that the amount of failure and/or success during play may have had an influence. To estimate whether a student might embrace or dismiss CCF at a specific moment in time during play, the following variables were used to formulate a logistic regression model with repeated measures and mixed effects:

- **Attending to CCF: *CCFatt*** (binary dependent variable). Classified as 1 if the student(s) had the message displayed on screen for more than 4.25 seconds, otherwise 0.
- **Condition for playing: *PlayCnd*** (categorical independent variable, fixed effects). Two conditions: Single and Pair.
- **Student Achievement Level: *AchLev*** (categorical independent variable, fixed effects). Two levels: Higher and MediumLower.
- **Task assessment category: *TaC*** (categorical independent variable, fixed effects). Categories: Almost Correct (75-80%), Quite Correct (60-74%), Quite Wrong (30-59%) and Very Wrong (<30%).
- **Trial number: *Nr*** (numerical independent variable, fixed effects). Only the first six unsuccessful trials for each mission were included in the dataset.
- **Student subject: *Subj*** (categorical independent variable, randomized effects per *PlayCnd*).
- **Interaction effects:** We also hypothesized interaction effects between *PlayCnd*, *AchLev* and *Nr*.

The following logistic model predicting CCF-attendance was then hypothesized:

$$\text{logit}(\text{CCFatt}) = \beta_0 + \left[\frac{\text{PlayCnd}}{\text{Subj}} \right] + \beta_1 \text{PlayCnd} + \beta_2 \text{AchLev} + \beta_3 \text{TaC} + \beta_4 \text{Nr} + \beta_6 \text{PlayCnd:AchLev} + \beta_7 \text{PlayCnd:Nr} + \beta_8 \text{AchLev:Nr} + \beta_9 \text{PlayCnd:AchLev:Nr}$$

The best mixed-effects logistic regression model was fitted to the data set using a step-wise-step forward procedure. All independent variables, interaction effects included, were found to be significant. First, as hypothesized, the probability for attending to CCF was lower for task trials with a higher number of errors (the task assessment categories 'Quite Wrong' and 'Very Wrong'). This pattern is consistent with findings in earlier studies using the same stimulus (Ternblad & Tärning, 2020). Second, even though the impact of the playing condition (*PlayCnd*) was small, we found an interesting interaction effect between this condition and the student achievement level (*AchLev*), revealing that while the probability for attending to the CCF (that is, not dismissing it before 4.25 seconds) were higher for lower-achieving students playing in pairs (compared to lower-achieving students playing individually), it was the opposite for higher-achieving students – who instead were more inclined to attend to the feedback when playing individually (compared to playing in pairs) (see Figure 2).

Another result – also in line with previous studies (Ternblad & Tärning, 2020) – is that the students tended to dismiss CCF to a larger extent after several attempts on the same task. However, a new finding in the present study was that the higher-achieving students playing in pairs were less sensitive towards this type of repeated failing (Figure 2). The final minimal adequate model for *CCFatt* performed significantly better than an intercept-only base line model ($\chi^2(10): 77.1, p < .001$), and had a reasonable fit (C-value: 0.79, Somers Dxy: 0.58). The model is visualized in Figure 2 and 3, and statistical details are presented in Table 2.

Figure 2

Probability of attending to CCF in relation to condition, achievement level and TaC

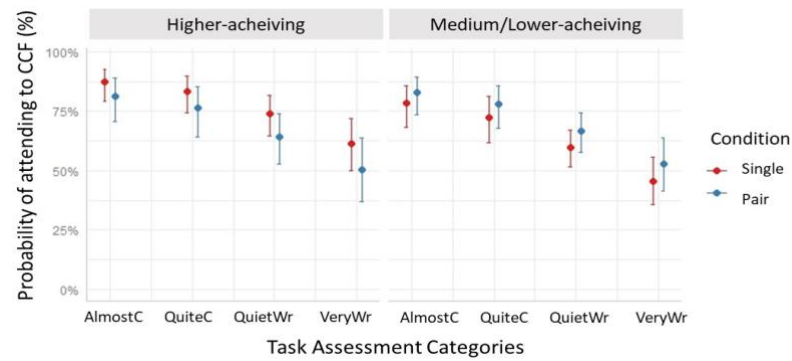


Figure 3

Probability of attending to CCF in relation to condition, achievement level and task trial

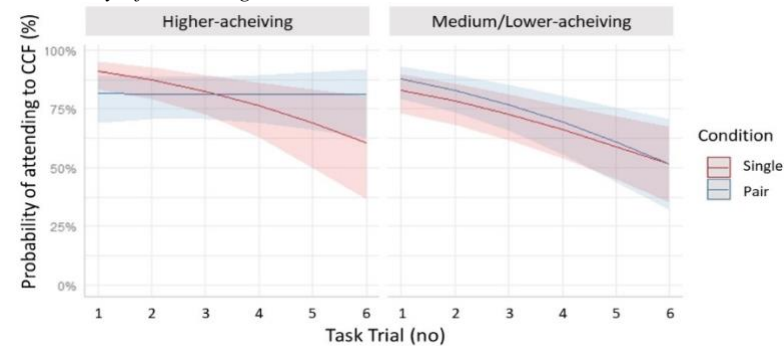


Table 2

Summary of the final minimal logistic mixed-effects regression model fitted to predict CCFatt

Random effects (Subject, N=106)		Variance		Std. Dev.	Corr	
Subject	Intercept		0.36	0.60		
	PlayCnd		0.60	0.77	-0.28	
Fixed Effects		Coeff.	Odds.Ratio	Std. Err.	z-value	Pr (> z)
(no of obs = 1043)						
Intercept		2.69	14.73	0.43	6.20	<.001 ***
PlayCnd[Pair]		-1.20	0.30	0.49	-2.43	<.05 *
AchLev[MediumLower]		-0.79	0.45	0.43	-1.83	0.07.
Nr		-0.38	0.68	0.12	-3.12	<.01 **
TaC[QuiteCorrect]		-0.31	0.73	0.29	-1.07	0.29 ns
TaC[QuiteWrong]		-0.89	0.41	0.24	-3.64	<.001 ***
TaC[VeryWrong]		-1.47	0.23	0.28	-5.21	<.001 ***
AchLev[MediumLower]:PlayCnd[Pair]		1.64	5.16	0.61	2.69	<.01 **
Nr:PlayCnd[Pair]		0.37	1.45	0.16	2.3	<.05 *
Nr:AchLev[MediumLower]		0.07	1.07	0.14	0.51	0.61 ns
Nr:AchLev[MediumLower]:PlayCnd[Pair]		-0.45	0.64	0.20	-2.23	<.05 *
Model statistics						Value
AIC						1309
C-value						0.79
Somer's Dxy						0.58
Likelihood ratio test						$\chi^2(10): 77.11, p<.001 ***$

Discussion

In this paper, we have explored the tendency for students to attend to feedback when using an educational game in history, either individually or in pairs. 5th and 6th grade students played the game individually for one lesson and in pairs for another lesson. The students' attentiveness towards feedback was evaluated via time stamped data logs for automatically displayed CCF-messages. By using the factor "attended towards the feedback or not" as a binary dependent variable, and the playing condition (pair/single) as one of the predicting variables in a logistic regression model, we can conclude that while medium/lower-achieving students were slightly more inclined to attend to the feedback when playing in pairs, this was not the case for the higher-achieving students. A possible reason for this result is that some of the tasks were quite easy, mitigating any positive effect of working in pairs for students with better performance and reading skills. This is in line with previous CL research (Nokes-Malach et al., 2012).

However, when higher-achieving students played in pairs, they were more inclined to attend to CCFs after repeated failing, revealing a consistent behavior independently of the number of trials on the same task. Since we know from previous studies that repeated failing might heavily influence students' inclination to accept and attend to feedback (Ternblad & Tärning, 2020), this is a new finding. But, as Nokes-Malach and others (2012) have pointed out, the strength with collaboration is dependent on the relation between task complexity and the competence of the group members. In our case, this positive effect for pair-working higher-achievers could be due to that the combined competence in these groups were optimal for some of the more difficult tasks, leading to better strategies and error-correction (see also Ross et al., 2004). All the other patterns – i. e., that repeated failures as well as many errors on a task often lead to a decline in attention to feedback – are consistent with earlier findings (Ternblad & Tärning, 2020). However, our own previous research on feedback acceptance has mainly targeted the impact of a teachable agent (a virtual character that is taught by the student, a so-called TA), evaluating if and how such an agent might scaffold students to embrace feedback more easily (Silvervarg et al., 2020; Ternblad & Tärning, 2020; Tärning et al., 2020). When comparing results from this study to earlier ones we can conclude that a TA seems to have a much stronger impact on feedback acceptance than if the students play in pairs. The only exception regards higher achieving students with repeated failures, who instead seem to encourage one another to attend to the CCF even if struggling (even if this assumption is slightly speculative).

There are, however, some limitations with the study at hand. First, letting an automatically delivered CCF-message being displayed on screen does not necessarily mean the same thing as "attending to it". It might have been better for the students to ask for the feedback themselves, or at least letting them respond to a yes/no-question whether they wanted the feedback or not. This would have made the interpretation of the data logs more straightforward. Second, the assumption that the pair-behavior derived from both students in a democratic way might not be totally true. And third, interpreting attention towards feedback in pairs as an indication of "collaborative" behavior that supports learning is hardly clear-cut. One might, for instance, argue that working in pairs automatically slows all interaction down, not only due to negotiations between pair members, but due to pure practicalities (such as competing over the screen etc.). However, since the behavioral patterns differ between higher achieving and medium/lower-achieving students this does not seem to be the case. We could therefore also emphasize that the most interesting findings in this study are the interaction effects, revealing changes in behavioral patterns between different types of students and/or between different degrees of failing.

The methodology in the experiment can also be discussed. At the same time, the study raises interesting questions for how one might investigate individual productive learning behaviors in cooperative settings. Advocators for CL often argue that collaboration may strengthen students' self-regulated learning (SRL), since students working in groups can help one-another to self-regulate (Schoor et al., 2015). But while individual approaches to SRL often focus on analyzing behavioral patterns, such as data traces etc. (see for instance Zimmerman, 2008), classical CL studies often rely on observations and analyses of conversations, searching for how learning unfolds through communication and shared understanding (Schoor et al., 2015). As mentioned in the background section, such data would have been interesting to add to this inquiry. For instance, by comparing utterances from students working in pairs (higher- versus medium/lower-achievers), differences in quantitative measures (such as attention towards CCF-messages or not) could perhaps be explained and better substantiated. However, without proposing one type of analysis or methodology to be better than another, we still propose data traces to be useful when studying differences between individual and collaborative learning, not at least when it comes to pinpointing if and how specific SRL-behaviors may change when a student moves from one learning situation to another.

Another interesting topic would be to investigate if and how CL would affect the students' reactions to feedback. That is, do working in pairs influence if or how they make use of the CCF when trying to solve the task again, and are students working in pairs more inclined to follow the hints presented to them in the CCF? Here, we know from previous studies that when students are working alone, very few of them actually 'do as they're told'

after receiving a feedback message (Silvervarg et al., 2020; Tärning et al., 2020). More commonly, students often try to revise a task by pure guessing, or by a trial-and-error-like behavior. If students working in pairs would be more inclined to, for example, gather more information (which in the present case involves making new time travels), this would also be a strong indicator for the benefits of collaborative learning. Finally, this study concerns a particular computer game, not specifically designed for collaboration. It would be of great interest to know if the feedback behaviors found in this study could be replicated with using other types of educational software.

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